

Day 3 — Complement to 1000

Module: Nikhila Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will find the complement of any 3-digit number to 1000 in under 4 seconds, and will recognise the same pattern extends to 10 000, 100 000, and beyond.

Materials

- Whiteboard
- Pair-drill sheets (20 three-digit numbers; second set with mixed 3- and 4-digit)
- Stopwatch
- Calculator (for self-verification only)

Vocabulary

- *pūṛaka* — complement (introduced Day 2; reuse)

Lesson flow

Warm-up (5 min)

- Sūtra chant 3 times.
- Quick fire: teacher calls 2-digit numbers; students chorus the complement to 100.
 - $73 \rightarrow 27$, $41 \rightarrow 59$, $28 \rightarrow 72$, $86 \rightarrow 14$, $95 \rightarrow 05$, $60 \rightarrow 40$, $17 \rightarrow 83$.

Core (15 min) — Three digits, same rule

1. **Pose:** “What’s the complement of 738 to 1000?”
2. Walk through:

$$\begin{array}{r} 1000 \\ - 738 \\ \hline \end{array}$$

- First digit (7): $9 - 7 = 2$
 - Second digit (3): $9 - 3 = 6$
 - Last digit (8): $10 - 8 = 2$
 - Answer: **262**.
 - Verify: $738 + 262 = 1000$. ✓
3. “Notice — same six-word rule. We just have one more ‘from 9’ step in the middle.”
 4. **The general pattern:**

For an n -digit number, the complement to 10^n is:

$$(9 - d_1), (9 - d_2), \dots, (9 - d_{n-1}), (10 - d_n)$$

where $d_1 \dots d_n$ are the digits left-to-right.

5. **Demo 4 more, with class chanting along:**

Number	Working	Complement	Check
419	9-4, 9-1, 10-9 = 5, 8, 1	581	419 + 581 = 1000 ✓
256	9-2, 9-5, 10-6 = 7, 4, 4	744	256 + 744 = 1000 ✓
803	9-8, 9-0, 10-3 = 1, 9, 7	197	803 + 197 = 1000 ✓
999	9-9, 9-9, 10-9 = 0, 0, 1	001 = 1	999 + 1 = 1000 ✓

6. **Quick extension to 10 000:** Complement of 2347 to 10 000?

- 9-2, 9-3, 9-4, 10-7 = **7653**.
- $2347 + 7653 = 10\ 000$. ✓
- “Same rule. Just more digits.”

Activity (15 min) — Speed drill

- Pair-drill: 20 three-digit numbers. Aim for all 20 in under 90 seconds.
- Self-check using a calculator: number + your complement should equal 1000.
- Second sheet: 10 four-digit numbers, complement to 10 000. Aim under 60 seconds for the set.

Wrap-up + Formative quiz (10 min)

- Formative quiz Part A from quizzes/formative.md — 5 questions on slips of paper, 5 min.
- Collect. Next class includes the result.

Student-facing content

Complement to 1000 (and beyond)

For any 3-digit number ABC: - First digit: **9 - A** - Second digit: **9 - B** - Last digit: **10 - C**

Same six-word rule. Same approach for 4-digit numbers (one extra “from 9”) and 5-digit numbers (two extra “from 9”s).

Examples: - 738 → 262 - 419 → 581 - 2347 → 7653 (complement to 10 000)

“Nikhilam navataścaramam daśataḥ — *all from nine and the last from ten.*” (Tirthaji, *Vedic Mathematics*, 1965, sūtra #2.) The “all” handles every digit except the last; “the last from ten” handles the last digit’s special role.

Homework

- Find the complement of each of these to 1000:
 - 156, 423, 789, 802, 64 (treat as 064), 7 (treat as 007), 555, 900.
- Find the complement of each of these to 10 000:
 - 3456, 1027, 8888, 75 (treat as 0075), 999.
- Time yourself. Total time goal: under 2 minutes.

Differentiation

- **Tier up:** Complement of 5-digit numbers to 100 000. Time goal: under 6 sec each.
- **Tier down:** Stay on 3-digit complements. The four-digit and five-digit cases come naturally once 3-digit is automatic.

Teacher notes

- The most common error: a number like 64 has only two digits; students forget the rule wants a 3-digit number, so they apply the rule to “64” (giving 36) instead of “064” (giving 936). Make sure they “pad with zeros” mentally.
- Another common error: the last digit being 0. For $800 \rightarrow$ complement to 1000: digit-wise gives $9-8, 9-0, 10-0 = 1, 9, 10$. The “10” carries. Easier reading: $800 + 200 = 1000$, so complement is 200. The rule still works ($1, 9, 10 \rightarrow 1, 10, 0 \rightarrow 200$), but it’s a place where the padding/carry interacts. Use this as a teachable moment; preview Day 6’s algebraic justification.
- Some students will sniff out a shortcut: “the complement just makes everything add to 9 except the last column which adds to 10.” Great — that *is* the rule, restated. Praise.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2).
- *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*, Nikhilam chapter — worked 3-digit and 4-digit examples.